

stirling's theorem notes

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1 Notes

Claim: $x^n/n! \leq e^x$ for all $x \geq 0$.

Proof: We induct on n . For $n = 0$, we have $1 \leq e^x$, true. Suppose the result for n . Then $\int_0^x t^n/n! dt \leq \int_0^x e^t dt$, or $x^{n+1}/(n+1)! \leq e^x - 1$, which is stronger than the desired result.

Specialize $x = n$, so $n^n/n! \leq e^n$, or $n! \geq (n/e)^n$. The other direct has $n! \leq n^n$.

Now use area test. $\log n! = \sum_{k=1}^n \log k \leq \int_1^{n+1} \log x dx = (n+1)(\log(n+1) - 1)$